

Work Session Documentation: Buck Converter Debug

Subsystem: Master Board Power Tree | Equipment: bench supply, DMM, oscilloscope, electronic load

Session Objective

The objective of this work session was to investigate why the initial pack-to-12 V buck converter stage did not produce a stable 12 V rail under expected operating conditions. The test focused on the inductor choice, output ripple behavior, and whether the converter was being pushed toward discontinuous conduction mode.

Raw Data and Observations

Test Condition	Observed Result	Immediate Note
Input set to 23.0 V, light load	12 V rail present but showed visible ripple on oscilloscope.	Converter was switching, but regulation quality was not acceptable.
Input increased toward 25.2 V	Ripple magnitude increased and downstream rails became less stable.	Worst-case full-charge voltage made the issue more visible.
Initial inductor value checked	Installed inductor was 4.7 uH.	Value was smaller than needed for the selected load and switching frequency.
Ripple-current calculation	Delta I _L calculated as approximately 2.67 A peak-to-peak.	Ripple was large compared with nominal load current.
Revised inductor calculation	10 uH reduced estimated ripple to approximately 1.25 A peak-to-peak.	Larger inductor should improve CCM margin.

What Was Tried

The buck converter was first tested using the original 4.7 uH inductor while varying the input voltage across the expected pack range. The team then calculated duty cycle and ripple current using the worst-case 25.2 V input condition. After identifying excessive ripple, the design was re-evaluated with a 10 uH inductor and the expected ripple current was recalculated.

What Happened

The original inductor value allowed too much ripple current. At 25.2 V input, 12 V output, and 500 kHz switching frequency, the calculated ripple was approximately 2.67 A. This was high enough that the converter could enter discontinuous conduction mode at light or moderate load. The revised 10 uH value reduced the estimated ripple to approximately 1.25 A, improving output stability and downstream regulation.

Inference

The failure was not caused by the buck IC being fundamentally unsuitable; it was caused by an inductor value that did not provide enough energy-storage margin for the operating point. The correct design inference was that power converters must be checked using worst-case input voltage, switching frequency, load current, ripple current, and conduction-mode limits before layout is finalized.